Emotion Analysis Using BRIGHTER + EthioEmo Datasets
Evaluating Data Augmentation Techniques for Multilabel Emotion Classification in Low-Resource African Languages

Marco Geral and Chisom Emekpo and Sean Maritz
Group 6

1 Introduction

Emotion analysis (EA) automatically identifies textual emotions, but African languages remain underrepresented due to linguistic diversity and data scarcity. Consequently, models struggle with multilabel classification, where a single text conveys multiple overlapping emotions. To mitigate this, this project investigates data augmentation, specifically back-translation and paraphrasing, to artificially expand training datasets. By evaluating these techniques alongside qualitative error analysis, this research seeks to improve multilabel emotion classification for low-resource African languages and build more inclusive, responsible NLP systems.

2 Background

Emotion analysis (EA) identifies specific, often overlapping emotional states (e.g., anger, joy) rather than basic positive or negative feelings, requiring multilabel classification for real-world accuracy (Plaza-del-Arco et al., 2024). However, the vast annotated data required by modern EA models creates a barrier for African languages, which remain heavily low-resource (Adebara and Abdul-Mageed, 2022). While recent Afrocentric efforts have advanced foundational NLP, complex tasks like emotion classification remain under-explored and rarely optimised for typological nuances like complex morphology and code-switching (Adebara, 2024).

To address data scarcity, researchers use data augmentation, specifically back-translation and paraphrasing, to artificially expand training sets. These approaches excel in low-resource environments by increasing lexical diversity and maintaining semantic fidelity, often outperforming AI models without prior examples (Radlinski et al., 2025). However, their application to multilabel emotion classification in African languages remains unexplored. Despite benchmarks like the EthioEmo dataset, the field is still vastly under-resourced. This project bridges this gap by evaluating back-translation and paraphrasing on multilabel emotion tasks for underrepresented African languages, providing essential insights to build inclusive NLP models.

3 Research Questions

1. How can data augmentation strategies (such as back-translation and paraphrasing) improve multilabel emotion classification for African languages?
2. How can existing NLP methods be adapted or applied effectively in low-resource African language contexts for emotion analysis?

4 Dataset and Data Description

4.1 Primary Dataset: BRIGHTER

This project uses the BRIGHTER dataset (Muhammad et al., 2025), a multilingual collection of approximately 100,000 emotion-annotated text instances across 28 languages. Each instance is labelled with six emotions (joy, sadness, anger, fear, surprise, disgust) and a neutral class, using a four-level intensity scale. The dataset supports multilabel classification, which is the focus of this study.

4.2 Selected Languages

Three African languages are selected: Afrikaans, Hausa, and Igbo. These were chosen based on data availability, societal relevance, and team familiarity. All have complete train, development, and test splits, enabling augmentation experiments. The languages represent distinct language families and together cover a large speaker base, supporting broader applicability of results.

4.3 Confirmed Availability

The dataset is publicly available on Hugging Face under a CC-BY 4.0 licence, and no additional data collection is required.

5 Approach and Methodology

5.1 Overview

This project establishes a systematic framework to investigate whether targeted data augmentation strategies mitigate performance degradation in multilabel emotion classification for low-resource African languages. Low-resource natural language processing (NLP) setups frequently suffer from high variance and overfitting during fine-tuning due to sparse semantic representations in the underlying embedding space.

The proposed methodology is structured across five primary execution phases: baseline initialisation, multi-path synthetic data generation using translation pivots, tokenisation and dynamic optimisation alignment, systematic comparative evaluation across isolated experimental conditions, and linguistic-informed qualitative error analysis.

5.2 Baseline Models and Mathematical Formulation

To establish a rigorous empirical performance lower-bound, we fine-tune a pretrained transformer architecture adapted for regional contexts: **AfroXLMR-base**. AfroXLMR is a language-adapted variant built upon the XLM-RoBERTa architecture, which was initially trained on 2.5TB of CommonCrawl data. AfroXLMR-base was post-pretrained on heavily curated, African-centric corpora covering multi-lingual web domains. This structural design leverages cross-lingual transfer learning uniquely tailored to the morphological nuances, syntactic variations, and local idioms of African languages.

5.2.1 Classification Head and Loss Function

Because emotion detection is framed as a multilabel classification task, where an individual text snippet can simultaneously manifest multiple non-mutually exclusive categories such as *joy*, *anger*, or *fear*, the standard single-label softmax layer is discarded. The output architecture consists of a dense linear projection layer mapping from the hidden state dimension H to a target dimension N, where $N = 6$ represents the individual target emotion classes.

Each raw unnormalised output logit z_i is mapped independently to a probability distribution using an element-wise Sigmoid activation function:

$$\sigma(z_i) = \frac{1}{1 + e^{-z_i}} \tag{1}$$

To optimise the network parameters θ without assuming mutual exclusivity among labels, we employ Binary Cross-Entropy (BCE) with Logits Loss. The total objective loss accumulated across a single instance is defined as the unweighted average of individual binary classification objectives over all N hidden targets:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \cdot \log(\sigma(z_i)) + (1 - y_i) \cdot \log(1 - \sigma(z_i))\right] \tag{2}$$

where $y_i \in \{0, 1\}$ represents the ground-truth binary indicator vector for emotion class i. During inference, a thresholding function converts continuous probability estimates to final hard labels such that $\hat{y}_i = 1$ if $\sigma(z_i) \geq \tau$, else 0, where the optimal decision boundary threshold is set at $\tau = 0.3$.

5.3 Data Augmentation Techniques

To alleviate textual data scarcity and regularize the neural representations, synthetic datasets are dynamically constructed via a sequence-to-sequence framework driven by the `NLLB-200` (No Language Left Behind) distilled 600M parameter model. Two main augmentation tracks are explored:

5.3.1 Back-Translation (Lexical Variation)

Source text in the target low-resource language (L_{src}) is translated to an intermediate high-resource pivot language (L_{pivot}) and subsequently translated back into the source format (L'_{src}). This pipeline introduces structural variation and lexical substitutions while keeping semantic boundaries invariant.

- **High-Resource Pivot:** Employs English (`eng_Latn`) to act as a primary translation pivot.
- **Regional Cross-Pivot:** Integrates French (`fra_Latn`) to leverage alternate translation paths found in regional African corpora.

5.3.2 Paraphrasing via Stochastic Decoding

To prevent deterministic translation collapse and generate richer semantic alterations, an advanced paraphrasing pipeline is constructed using intermediate pivots combined with stochastic inference. Rather than executing a standard greedy or beam search, the reverse generation phase employs nucleus sampling (top-p) alongside top-k filtering. The probability distribution for token selection at step t over the truncated vocabulary $V^{(p)}$ is formulated as:

$$P(x_t \mid x_{<t}) = \frac{\exp(z_t/T)}{\sum_{j \in V^{(p)}} \exp(z_j/T)} \quad (3)$$

By keeping $k = 50$ and $p = 0.95$, the decoding engine samples creatively from the highest probability mass, introducing vocabulary synonyms and idiom paraphrases while preserving original text labels.

5.4 Experimental Conditions

To isolate the exact downstream performance gains of individual data strategies versus dense combined setups, the model is exposed to four distinct data boundaries:

- **Condition A: Baseline (1x Data):** The control benchmark environment using only the human-annotated training dataset ($2,145$ samples).
- **Condition B: Back-Translation Only (2x Data):** The baseline data appended with English-pivoted back-translations ($4,290$ total samples).
- **Condition C: Paraphrasing Only (2x Data):** The baseline data appended with stochastic-sampled paraphrased permutations ($4,290$ total samples).
- **Condition D: Combined Pipeline (4x Data):** A dense stacked training set pooling Original data, English back-translations, French back-translations, and stochastic paraphrases ($8,580$ total samples).

5.5 Optimization and Hyperparameters

All model parameters are fine-tuned using the `AdamW` optimiser across a fixed schedule of 10 epochs. To maintain training stability and prevent catastrophic forgetting as data volume scales across conditions, learning rates are dynamically adjusted. Conditions A and B use an initial learning rate of 5×10^{-5}, whereas the expanded sample spaces of Conditions C and D utilise a conservative 3×10^{-5} tied to a cosine learning rate scheduler. Weight decay is uniformly held at 0.01, with a per-device batch size of 16. The primary evaluation metric tracked is the **Macro-Averaged F1-Score** to ensure classification quality is weighed evenly across minority and majority emotion states.

5.6 Error Analysis

To look beyond aggregate quantitative trends, a post-hoc qualitative error analysis is integrated into the workflow. Predictions from the optimal models are extracted and mapped into multi-label confusion matrices. Misclassifications are manually isolated and scrutinized across three core linguistic dimensions:

1. **Emotion Overlap and Ambiguity:** Investigating instances where high valence affinity (e.g., *Sadness* vs. *Anger*) blurs individual class activation boundaries.
2. **Linguistic Complexity:** Tracking how colloquial structures, local code-switching, and regional expressions disrupt the tokenisation patterns of the AfroXLMR vocabulary.
3. **Class Imbalance Distortions:** Assessing if lower frequency labels are regularly dominated by high-density label predictions despite the synthetic data expansions.

6 Experiments and Results

results will go here

7 Reflections

reflections will go here

8 Conclusion

conclusion will go here

References

A Example Appendix

This is an appendix.

Table 1: Structural Specifications of Experimental Dataset Configurations

Condition	**Augmentation Strategy**	**Multiplier**	**Total Training Samples**
Condition A	Baseline Control (No Augmentation)	1×	2,145
Condition B	Back-Translation (`eng_Latn`)	2×	4,290
Condition C	Stochastic Paraphrasing Only	2×	4,290
Condition D	Combined Ensemble (BT + Para)	4×	8,580